

Nut consumption and risk of type 2 diabetes, cardiovascular disease, and all-cause mortality: a systematic review and meta-analysis.

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BACKGROUND: Epidemiologic studies have shown inverse associations between nut consumption and diabetes, cardiovascular disease (CVD), and all-cause mortality, but results have not been consistent. OBJECTIVE: We assessed the relation between nut intake and incidence of type 2 diabetes, CVD, and all-cause mortality. DESIGN: We searched PubMed and EMBASE for all prospective cohort studies published up to March 2013 with RRs and 95% CIs for outcomes of interest. A random-effects model was used to pool risk estimates across studies. RESULTS: In 31 reports from 18 prospective studies, there were 12,655 type 2 diabetes, 8862 CVD, 6623 ischemic heart disease (IHD), 6487 stroke, and 48,818 mortality cases. The RR for each incremental serving per day of nut intake was 0.80 (95% CI: 0.69, 0.94) for type 2 diabetes without adjustment for body mass index; with adjustment, the association was attenuated [RR: 1.03; 95% CI: 0.91, 1.16; NS]. In the multivariable-adjusted model, pooled RRs (95% CIs) for each serving per day of nut consumption were 0.72 (0.64, 0.81) for IHD, 0.71 (0.59, 0.85) for CVD, and 0.83 (0.76, 0.91) for all-cause mortality. Pooled RRs (95% CIs) for the comparison of extreme quantiles of nut intake were 1.00 (0.84, 1.19; NS) for type 2 diabetes, 0.66 (0.55, 0.78) for IHD, 0.70 (0.60, 0.81) for CVD, 0.91 (0.81, 1.02; NS) for stroke, and 0.85 (0.79, 0.91) for all-cause mortality. CONCLUSIONS: Our meta-analysis indicates that nut intake is inversely associated with IHD, overall CVD, and all-cause mortality but not significantly associated with diabetes and stroke. The inverse association between the consumption of nuts and diabetes was attenuated after adjustment for body mass index. These findings support recommendations to include nuts as part of a healthy dietary pattern for the prevention of chronic diseases. © 2014 American Society for Nutrition.